1 Lecture 08 LRM Short Answer Questions

2 Question #1

Let y be a random variable with $\mathrm{E}(y) = \mu$. The expected squared error when a is used as a prediction of y is

$$\mathrm{E}\left[(y-a)^2\right].$$

Show that the value of a that minimizes expected squared prediction error is the expected value of y.

3 Question #2

Let y be a random variable, and let

$$m(x) = \mathrm{E}(y \mid x)$$

denote the conditional expectation function for y.

a. Define

$$y = m(x) + u.$$

Show that

$$\mathrm{E}(u \mid x) = 0.$$

b. Let $h(x)$ denote any function of x. Using the result from part (a), show that

$$\mathrm{E}(h(x)u) = 0.$$

c. The expected squared error when the function $g(x)$ is used as a prediction of y is

$$\mathrm{E}\left[(y-g(x))^2\right].$$

Using the result from part (b), show that the function $g(x)$ that minimizes expected squared prediction error is the conditional expectation function $m(x)$.

4 Question #3

The OLS estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$.

The OLS solution satisfies the first-order conditions

$$-2\sum_{i=1}^{n} x_{ik}\left(y_i - x_{i1}b_1 - x_{i2}b_2 - \cdots - x_{iK}b_K\right) = 0,$$

for each $k = 1, 2, \ldots, K$.

Define the OLS prediction error as

$$e_i = y_i - x_i'b.$$

a. Suppose $x_{i1} = 1$ for all i, so that the first column of X is a constant. Show that

$$\sum_{i=1}^{n} e_i = 0.$$

b. Suppose $x_{i1} = 1$ for all i. Show that

$$\bar{y} = \bar{x}'b,$$

where

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

c. Show that the OLS prediction errors have zero sample covariance with each explanatory variable:

$$\widehat{\mathrm{Cov}}(x_k, e) = 0, \qquad k = 1, 2, \ldots, K.$$

For this problem, define

$$\widehat{\mathrm{Cov}}(x_k, e) = \frac{1}{n}\sum_{i=1}^{n}(x_{ik} - \bar{x}_k)(e_i - \bar{e}),$$

where

$$\bar{x}_k = \frac{1}{n}\sum_{i=1}^{n} x_{ik}, \qquad \bar{e} = \frac{1}{n}\sum_{i=1}^{n} e_i.$$

5 Question #4

The least squares estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$.

Consider the bivariate prediction equation

$$\hat{w} = b_1 + b_2 e.$$

For concreteness, let w denote wage and let e denote education. In this case, X is an $n \times 2$ matrix with a first column of ones and a second column containing the observed values of education:

$$X = \begin{bmatrix} 1 & e_1 \\ 1 & e_2 \\ \vdots & \vdots \\ 1 & e_n \end{bmatrix}, \qquad Y = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}.$$

Define the following sample averages:

$$\bar{e} = \frac{1}{n}\sum_{i=1}^{n} e_i, \qquad \overline{ee} = \frac{1}{n}\sum_{i=1}^{n} e_i^2,$$

$$\bar{w} = \frac{1}{n}\sum_{i=1}^{n} w_i, \qquad \overline{ew} = \frac{1}{n}\sum_{i=1}^{n} e_i w_i.$$

a. Starting from $b = (X'X)^{-1}X'Y$, show that the least squares estimator can be written as

$$b = \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}.$$

In your derivation, explicitly show the matrix multiplication using the elements of e and w that produce $X'X$ and $X'Y$, and then use the definitions of the sample averages above to obtain the final expression.

b. Use matrix algebra to show that the slope estimator can be written as

$$b_2 = \frac{\widehat{\text{Cov}}(e, w)}{\widehat{\text{Var}}(e)}.$$

For this problem, define the sample covariance and variance as

$$\widehat{\text{Cov}}(e, w) = \overline{ew} - \bar{e}\bar{w}, \qquad \widehat{\text{Var}}(e) = \overline{ee} - \bar{e}^{\,2}.$$